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Palaeomagnetism of Permian Redbeds from the South-western United States

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Summary

Samples were collected from Permian redbeds of Guadalupian to Ochoan age at 61 sites in the south-western U.S.A. Upon thermal demagnetization, a stable primary component of magnetization could be isolated in most of the samples. Because of the elimination of most of the secondary component of magnetization, the grouping of the virtual pole positions is better than previous results have suggested. There is no apparent chronological migration of the pole position through Permian time. Measurement has shown that although most of the Permian rocks are reversely magnetized, the rocks from four formations appear to have stable primary components approximately anti-parallel to the directions of magnetization of the reversed rocks. These rocks are of middle Leonardian, late Guadalupian, and middle and late Ochoan ages and it is suggested that they represent three, and possibly four, normal polarity events during the Permian.

Introduction

It is something of a paradox that rocks of Permian age in the western United States, some of the most suitable for palaeomagnetic work, are not better known. At the root of the situation is that much of the sampling and measurement was carried out before magnetic cleaning became a routine process (Graham 1955; Runcorn 1955, 1956; Doell 1955; Collinson & Runcorn, 1960). Subsequently, McMahon & Strangway (1968a, b) used A.C. Stability tests and later, after the present project had begun, Farrell & May (1969) reported the results of thermal demagnetization of samples from several different formations on the Colorado Plateau.

The present study was designed as an attempt to sample the Permian at controlled stratigraphic intervals using the correlation chart of Dunbar *et al.* (1960) and to apply standard thermal demagnetization to all samples. This, it was hoped, would provide the beginning at least of a 'type palaeomagnetic' sequence. Samples were obtained from 61 sites in Permian rocks in north-western and western Oklahoma, the Monument Valley region of Utah, the Grand Canyon, Oak Creek Canyon, and central and south-eastern New Mexico (see Fig. 1). The work was carried out with the support of the National Science Foundation Grant GA–1425.

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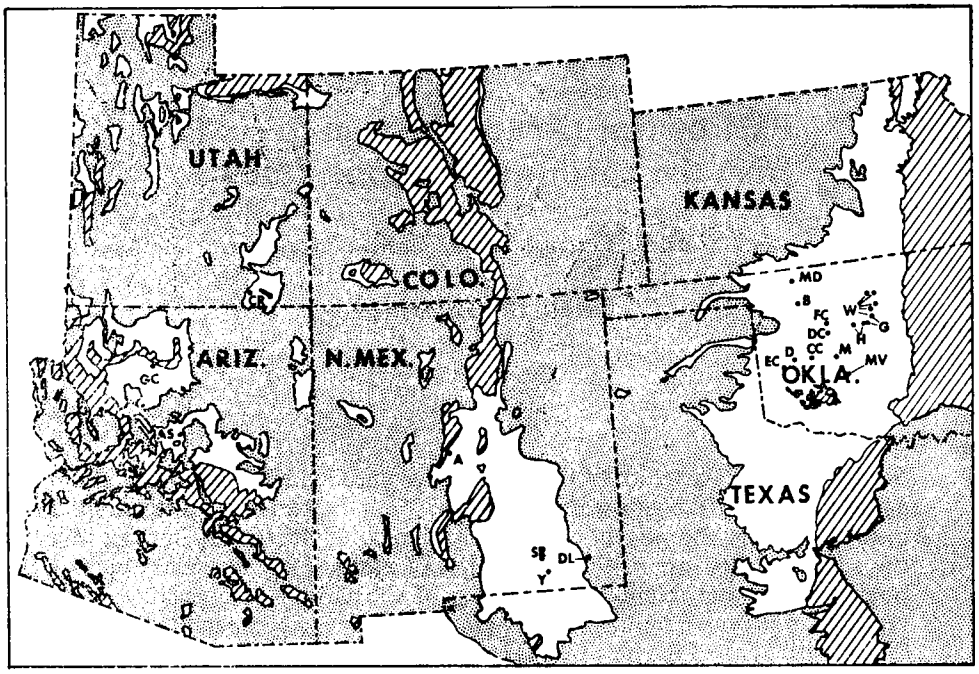


FIG 1. Map of outcropping Permian rocks in south-western United States and locations of collecting sites. The unmarked regions on the map are the areas where Permian rocks are exposed at the surface. Pre-Permian rocks at the surface are indicated by diagonal lines and post-Permian by stippled areas. The code for sampling sites is the same as in Table 1 except for at Comb Ridge, Utah and Grand Canyon, Arizona where samples were taken from several different units at one location. They have been designated on the map CR and GC respectively.

General geology

Sedimentary rocks of Permian age are exposed over a wide area in a belt running from south-eastern Nebraska through Kansas and Oklahoma and into north central Texas, the Colorado Plateau, and the West Texas–New Mexico area. Every mode of sedimentation from deep-water marine to continental can be found, but of particular interest are the redbed sequences. The palaeographical situation appears to have been a lowland region bordering an epi-continental sea. The oscillation of the shoreline led to the intercalation of marine horizons, by means of which the largely unfossiliferous redbeds can be tied to the standard marine succession. The stratigraphic chart (Fig. 2) is taken from Dunbar *et al.* (1960) and indicates the formations sampled for the investigation. Detailed references to the regions and descriptions of the formations in question can be found in the references cited by Dunbar.

The origin of redbeds and, linked to this, the origin of the remanent magnetization has long been a vexed question. Most of the formations examined in this paper belong to the 'Coastal Plains–Tidal Flat Redbeds on Cratons' categories of van Houten (1961), and it is generally assumed, and will be assumed here, that the origin of these redbeds was contemporaneous with deposition or occurred within a geologically short time thereafter. However, cognizance must be taken of the work of Walker (1967) which suggests that the acquisition of the red coloration may be appreciably delayed.

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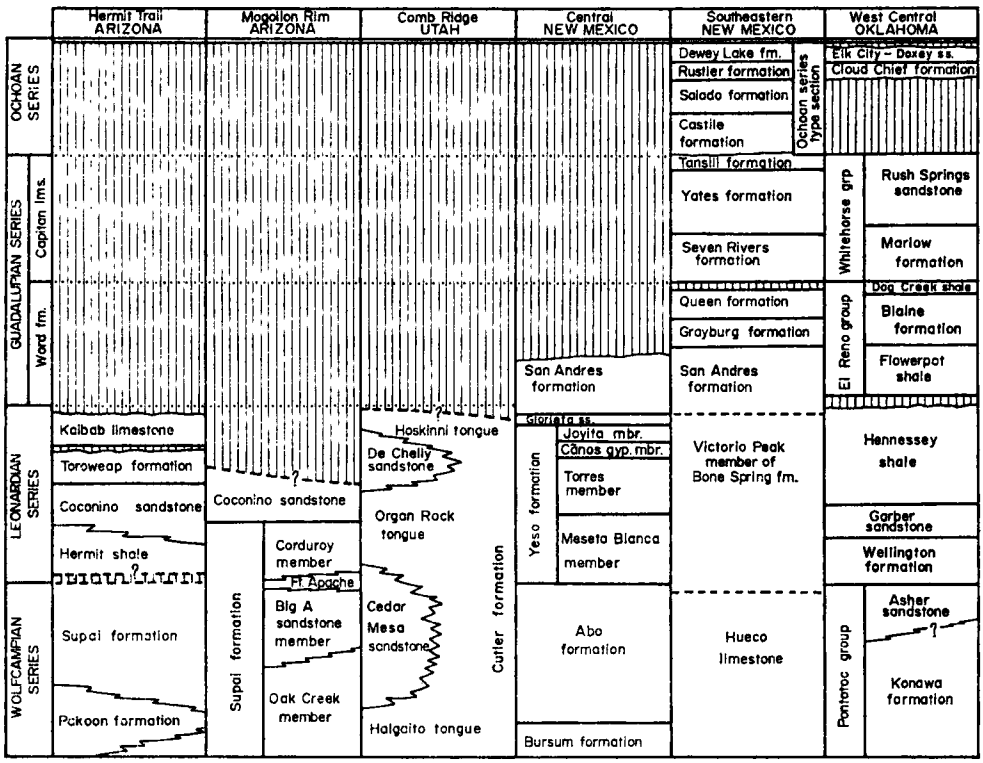


FIG. 2. Permian stratigraphic correlation chart (from Dunbar *et al.* 1960) showing sections for the regions collected.

Palaeomagnetic techniques

It is not easy to decide upon the best sampling technique for sedimentary rocks, since unlike lavas, there is no *a priori* control over the duration of the time represented, except in rare instances (e.g. varved clays). In consequence, it was decided to sample with approximately the same density, about six samples per 3–5 m stratigraphic interval with the maximum lateral spread practicable (up to about 50 m), and to regard this as a site. Therefore, site means have the same composition, and as they represent roughly the same stratigraphic interval, although no safe conclusions can be drawn about the time interval represented it seems not unreasonable in the absence of any obvious sedimentary breaks to assume that the time intervals may be comparable. Where thick sections were exposed and were readily accessible, collecting sites were established every 20 m through the exposed section. In computing a mean direction for a given formation, samples were given unit weight irrespective of the number of sites. The effect of this was to give smaller circles of confidence than would have been the case using site mean directions. However as in only two cases were there results from more than four sites per formation, it was not possible to base formation results on site means.

At most sites cores were obtained. At others where the material was too friable or did not hold together during drilling because the rock disintegrated in the drilling fluid, block samples were taken. At yet other sites, the rock was unsuitable for sampling by either technique because the matrial was too friable, too finely jointed, too fissile or too soluble. Thus, it proved impossible to obtain any samples from the Rush Springs formation of Oklahoma and the Yeso formation of New Mexico

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and severely restricted the sampling to one or two sites in the following formations: the Flowerpot shale, the Dog Creek shale, the Blaine formation, the Marlow formation, the Yates formation, and the Dewey Lake redbeds.

In general, intensities of magnetization were low. Measurements were made using a Foster-type fluxgate spinner (Foster 1966) with a maximum sensitivity of about 2×10^{-7} emu cm^{-3}. The samples proved to have high covercivities, and alternating fields in excess of 1000 oersteds had little effect. Magnetic cleaning was routinely carried out by progressive thermal demagnetization of two samples per site in steps of 50 °C up to 600 °C. The remainder of the samples were subsequently demagnetized in only two or three stages (400, 500 and 600 °C). The furnace was housed in a non-magnetic room of the type described by Patton & Fitch (1962), and a pair of auxiliary coils was used to further reduce the residual field to below 50 γ, usually the residuals measured were on the order to 20–30 γ or less. Very few samples did not possess appreciable secondary components of magnetization. This is illustrated in Fig. 3(b) which are plots of the changes in declination and inclination with increasing temperature against normalized values of the intensity.

Site mean directions and Fisher statistics were computed on a Wang model 373 computer using a program modified from Grommé (1965). The least scatter (max k) and the smallest circle of confidence was used to select what was regarded as the best representation of the original direction of magnetization at the time of formation of the rock (or a geologically insignificant time thereafter). Of the 61 sites sampled,

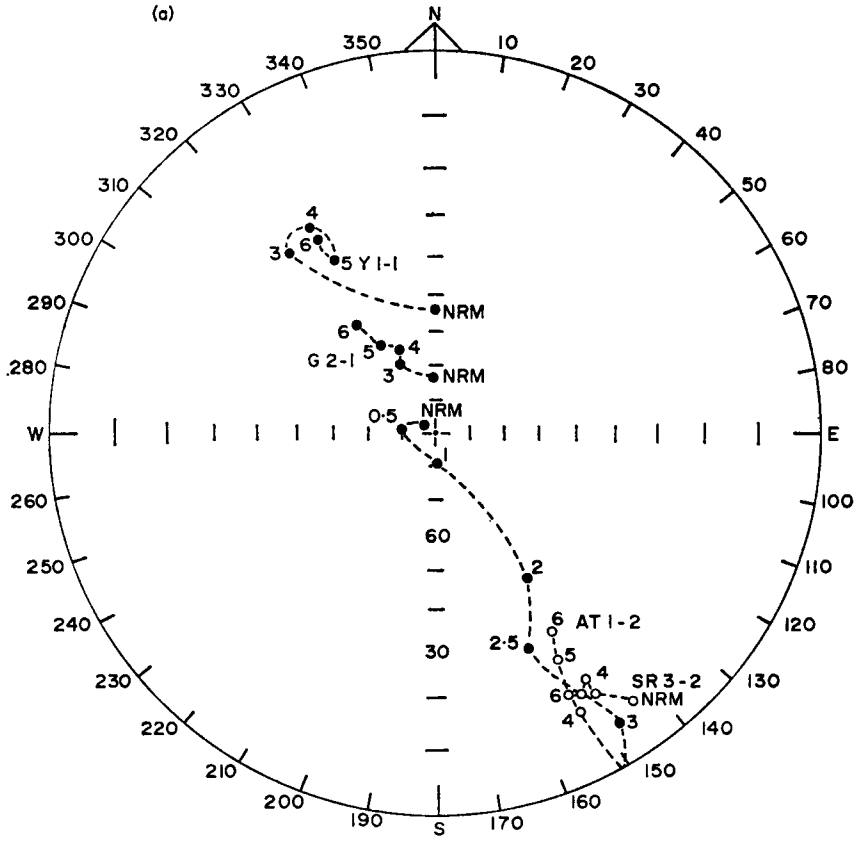


Fig. 3(a) Stereo plot showing demagnetization curves for four samples (temperatures given in 100's of degrees Celsius). Closed circles indicate positive inclinations and open circles denote negative inclinations.

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there still appears to be a significant secondary component after demagnetization at 500 or 600 °C in the samples from 18 of the sites. Arbitrarily, sites at which the circle of confidence exceeds 25° have been disregarded.

The site mean directions and relevant statistics are given in Table 1. For those formations in which more than one site was collected, a mean for the formation was also computed, and they are given separately in Table 2 together with the results from the previous investigations.

Results

Wolfcampian formations

Rocks of Wolfcampian age were collected at four localities: (1) the Cutler formation from Comb Ridge, Utah in the Monument Valley region, (2) the Supai formation in Oak Creek Canyon, Arizona, (3) the Supai formation from along the Hermit Trail in Grand Canyon, Arizona, and (4) the Abo formation in Central New Mexico.

The Cutler formation at Comb Ridge, Utah is divided into four units which are, from oldest to youngest: the Halgaito tongue, the Cedar Mesa sandstone member, the Organ Rock tongue, and the de Chelly sandstone member. According to Dunbar *et al.* (1960), the Wolfcampian–Leonardian boundary lies somewhere in the Cedar Mesa member. The Halgaito tongue is composed of reddish-brown, shaly siltstones. The Cedar Mesa sandstone is an orangish-brown to light red, cross-bedded, fine to-medium-grained sandstone with thin beds of grey, limey siltstones. Three sites

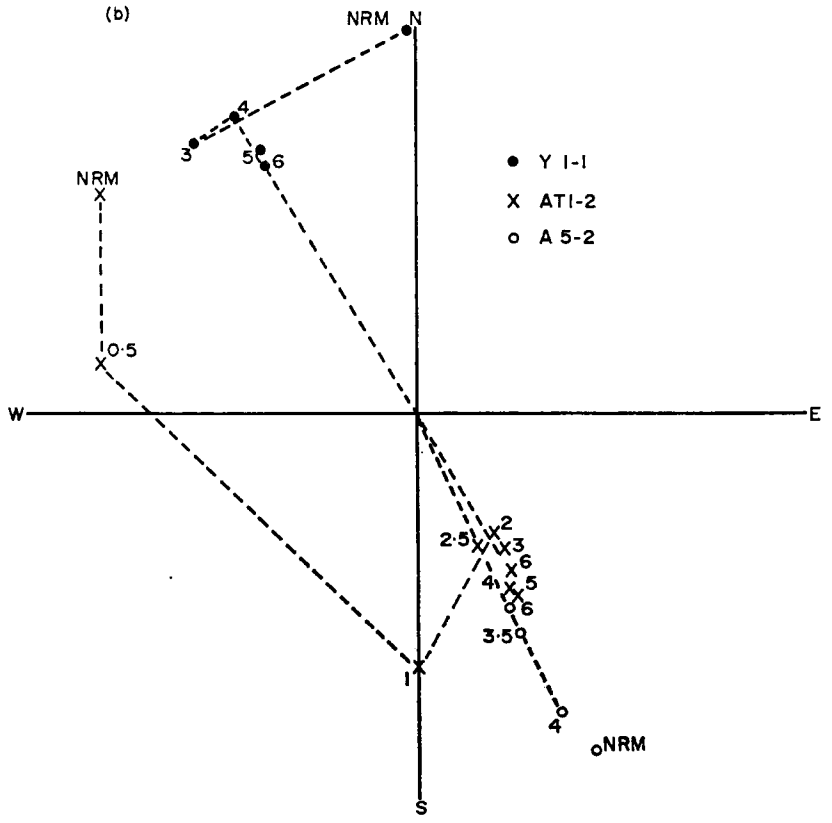


Fig. 3(b) Demagnetization curves showing changes in declination and normalized intensity (J/J_0) as a function of temperature.

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Table 1

Site Mean Directions, Fisher Statistics and Northern Hemisphere Virtual Pole Positions

The table lists under sampling the number of samples used in the calculation of the mean (*a*) and the total numbers collected (*b*). The temperature used in magnetic cleaning is also quoted.

	Magnetization Sampling					Virtual pole		
Site	*D*	*I*	a/b	*k*	α_{95}	Long. °E	Lat. °N	Temp. (°C)
Wolfcampian results (by site)								
UH2	128	−2	6/6	13	19	135	30	500
UCM1	139	+12	5/5	15	21	120	32	500
UCM2	142	+13	5/6	56	10	116	34	400
AS1	144	+12	5/6	21	17	116	37	500
AS2	168	−7	6/6	63	8	101	57	500
AS3	173	0	6/6	86	7	80	54	600
AS4	174	−3	6/6	149	5	78	56	500
AS5	168	−1	6/6	54	9	89	54	600
AS6	162	0	6/6	70	8	98	52	600
AS7	157	+19	4/6	26	18	99	41	600
AS8	160	−5	6/6	68	8	103	53	500
AS10	156	+10	5/6	15	20	102	44	500
GS2	152	+13	4/5	14	25	105	40	500
A1	147	−5	4/6	188	7	126	46	400
A4	142	−2	5/6	15	21	128	41	600
A5	159	+10	6/6	103	7	104	46	500
A6	157	−3	6/6	36	11	112	50	500
A7	149	+10	6/6	45	10	117	41	500
A8	157	+3	6/6	175	5	109	48	400
Leonardian results								
W2	158	−7	5/6	301	4	119	51	500
W5	160	−5	6/6	279	4	116	51	500
G2	348	+41	7/7	50	9	125	74	600
H2	157	−10	5/5	22	17	122	53	500
UO1	145	+26	4/6	16	24	108	29	500
UO2	150	+13	4/6	98	9	108	38	500
UO3	118	+23	4/5	16	23	132	14	500
AH1	158	+32	6/6	40	11	093	33	400
AT1	154	−20	6/6	78	10	117	55	620
Guadalupian								
B	160	−5	3/3	17	31	115	51	600
DC	156	−1	4/4	40	15	119	48	500
M	159	−11	7/7	72	7	119	54	500
SR1	148	−8	3/6	54	17	129	49	500
SR2	142	−14	5/6	18	19	138	46	400
SR3	152	−14	5/6	68	9	127	54	400
Y1	331	+16	6/6	17	17	130	54	400
Ochoan								
CC	357	+33	6/6	33	12	090	72	500
D	153	−2	5/6	69	9	123	47	500
EC1	152	−8	6/6	81	7	126	49	400
EC2	154	−12	6/6	309	4	126	53	400
EC3	164	−12	6/6	25	14	112	57	400
EC4	157	−8	4/5	338	5	120	52	500
DL	336	+5	4/6	220	6	119	52	500

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Table 2

Stratigraphical summary of Paleomagnetic data

The mean direction of each formation reported is based upon giving each sample unit weight irrespective of the number of sites. Where no α_{95} is given, data are taken from Table 1.

	Site	Samples	D	I	α_{95}	k	Virtual pole	
Ochoan							°E	°N
Dewey Lake fm.	1	4	336	+5	—	—	119	52
Elk City fm.	4	22	157	−10	4	56	122	53
Doxey fm.	1	5	153	−2	—	—	123	47
Cloud Chief fm.	1	6	357	+33	—	—	90	72
Guadalupian								
Yates fm.	1	6	331	+16	—	—	130	54
Seven Rivers fm.	3	13	147	−13	7	33	133	50
Marlow fm.	1	7	159	−11	—	—	119	54
Dog Creek fm.	1	4	156	−1	—	—	119	48
Blaine fm.	1	3	160	−5	—	—	115	51
Leonardian								
Hennessey	1	5	157	−10	—	—	122	53
Toroweap-Alpha mbr.	1	6	154	−20	—	—	117	55
Toroweap-Alpha mbr.*	—	11	157	+3	7	36	103	47
Garber fm.	1	7	348	+41	—	—	125	74
Wellington fm.	2	11	159	−6	—	—	118	51
Hermit	1	6	158	+32	—	—	93	33
Hermit*	—	4	161	−7	8	76	101	53
Cutler–Organ Rock mbr.	3	12	138	+21	12	14	117	28
Wolfcampian								
Supai fm. (Osk Creek Canyon)	10	50	163	+1	8	40	96	51
Supai fm. (Hermit Trail)	1	5	152	+13	—	—	105	40
Supai fm.**			151	+11	10	35	110	40
Cutler fm.-Halgaito-Cedar Mesa mbr.	3	16	134	+9	9	17	126	30
Cutler fm.-Halgaito mbr.	—	14	151	−11	6	54	117	49
Abo fm.	6	33	152	+2	8	77	116	46
Dunkard fm.** *	9	57	163	+8	5	162	123	44

*Farrell & May 1969
**Irving 1964
***Helsley 1965

were collected in the Halgaito tongue. From the scatter of the measured directions, it appears that even after demagnetization at 500 °C there is still a sizeable secondary component; however, further demagnetization reduces the remanent magnetization below the limit of measurement. The results from the two sites in the Cedar Mesa member are less scattered than those for the Halgaito tongue. The between-site difference is very small as can be seen from the following mean directions of magnetization:

	D	I	k	α_{95}	Long.	Lat.	Temp. °C
UCM1	139	+12	15	21	120° E	32° N	500
UCM2	142	+13	56	10	116° E	34° N	400

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The Supai formation in Oak Street Canyon is composed mainly of red, cross-bedded sandstones with one limestone member, the Fort Apache limestone. The base of the Supai is Pennsylvanian, and the top extends into the Guadalupian. Ten sites were collected in the Supai of Oak Creek Canyon, five from each two locations. The stratigraphic distance between each site was about 20 m depending upon the topography, exposure, accessibility, etc. The samples were taken up to near the base of the Coconino sandstone. These results give site directions and pole positions consistently 10–20° different from the results from other formations of Wolfcampian age although the differences are not significant at the 95 per cent confidence level (i.e. the circles of confidence overlap). The difference is believed to be due to a compass error, for according to Noltimier (personal communication), there is a local magnetic anomaly resulting from the lavas capping the canyon.

In the Grand Canyon the Esplanade member of the Supai formation was sampled at only two sites along the Hermit Trail. At this location the Esplanade member consists of 90 m of alternating beds of red, soft sandstone and buff, massive cross-bedded sandstone passing downward into more massive beds of reddish, purplish or buff cross-bedded sandstone. The intensities of magnetization were low, and the results which are very highly scattered even after demagnetization at 400 °C are as follows:

	D	I	k	α_{95}	Long.	Lat.	Temp. °C
GS1	137	+48	7	28	106° E	14° N	400
GS2	152	+13	14	25	105° E	40° N	500

It is apparent that at neither site is all of the secondary component eliminated. At site GS2 although the directions are more characteristic of previous Wolfcampian results, it still appears that not all of the secondary component has been eliminated and upon heating of the samples to 500 °C, the intensities are so low that measurement is extremely difficult, if not impossible.

The Abo formation was sampled at eight sites along U.S. highway 60 from its base just west of Scholle, New Mexico to the top at the top of the hill just east of the road to Abo Ruins. The Abo formation is a succession of dark-red sandstones and siltstones, in part conglomeratic (Needham & Bates 1943). The directions obtained from the individual sites are fairly consistent (Table 1). The one exception to this being the results from site A2 at which the mean inclination is 25–40° higher than for any other site in the formation and has, on this basis, been rejected. It is assumed that this is probably the result of secondary magnetization that has not been eliminated even upon heating to 640 °C. The results from site A3 have also been rejected because the radius of the circle of confidence exceeded 25°. The mean directions, Fisher statistics, and palaeopole positions were computed for the whole formation, giving the site means unit weight and a second time giving each sample unit weight. The results were as follows:

	D	I	k	α_{95}	Long.	Lat.
Sites	152	+2	77	8	116° E	46° N
Samples	153	+3	32	5	115° E	46° N

Leonardian formations

Rocks of Leonardian age were collected at four localities: north central and north-western Oklahoma, and Comb Ridge area of south-eastern Utah, Oak Creek Canyon and Grand Canyon. The Yeso formation of south central New Mexico could not be sampled at the locality visited near Socorro, New Mexico.

In Oklahoma three formations (see Fig. 2), the Wellington formation, the Garber sandstone, and the Hennessey shale, were sampled. In this region the Wellington formation consists of about 200 m of alternating beds of red shale and soft, red

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sandstone. The Garber formation is composed of a series of red clay shales, red sandy shales, and massive red sandstones. The Hennessey shale consists primarily of deep-red clay shale with interbedded greenish-gray siltstone and orange-brown sandstone and siltstone beds.

The Wellington formation was sampled at five sites only two of which yield consistent results (Table 1). The mean directions of magnetization from these two sites are similar:

	D	I	k	α_{95}	Long.	Lat.	Temp. °C
W2	158	−7	301	4	119° E	51° N	500
W5	160	−5	279	4	116° E	51° N	500

The results from the remaining three sites were still widely scattered even after demagnetization.

The Garber sandstone was sampled at only two sites. Although somewhat scattered directions of magnetization were obtained, the mean direction is some 130° removed from the mean of the Leonardian sites. It is considered that this represents a real even of normal polarity. Because of the interest of a normal event at this time during the Permian, a demagnetization diagram is given in Fig. 3. The site mean directions are the following:

	D	I	k	α_{95}	Long.	Lat.	Temp. °C
G2	348	+41	50	9	125° E	74° N	600

Two sites were collected from the Hennessey shale, the first of which gave the scattered reversed directions with a mean inclination suggesting that not all of the secondary component has been eliminated. The directions at site H2 are also reversed and are more tightly grouped:

	D	I	k	α_{95}	Long.	Lat.	Temp. °C
H2	157	−10	22	17	122° E	53° N	500

At Comb Ridge, Utah the upper part of the Cutler formation lies within the Leonardian, and that member of it sampled, the Organ Rock, is included here. At this location the Organ Rock tongue consists of about 200 m of brownish- to orangish-red, fine-grained sandstone and sandy mudstones. Samples were collected at three sites along the south side of state highway 44 near the junction with the Mormon Trail Road. Of the three sites, only one, UO2, has a high k value ($k = 98$) with a correspondingly low radius for the circle of 95 per cent confidence ($\alpha_{95} = 9°$). To obtain mean directions for the Organ Rock member, all of the individual values of D and I, except for six, were averaged giving each sample unit weight. Those six were rejected on the basis of the obvious influence of secondary components not eliminated in the thermal demagnetization process (i.e. all were rejected because the measured declinations were streaked back towards normal directions). The palaeo-pole position obtained (116° E, 34° N) is in the same area as most of the other pole positions, but for the Organ Rock member, as well as for the Cedar Mesa member, the palaeolatitude is slightly lower than for most of the other formations. This is, undoubtedly, the result of uneliminated secondary magnetization.

Rocks of Leonardian age were sampled from two formations, the Hermit shale and the Alpha member of the Toroweap formation, along the Hermit Trail in Grand Canyon, Arizona. At this site the shale is composed of 80–95 m of brick-red, sandy shale and fine-grained sandstone. The Alpha member of the Toroweap formation is a 50 m thick unit of yellow, white, and red fine-grained, crumbly sandstones interbedded with ledge-forming limestones. Two sites were collected in the Hermit shale, one 12 m above the lower contact with Supai formation and the other 40 m above the

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first site. The Alpha member of the Toroweap formation was collected at only one site. The results from the Hermit shale are better for the first site than the second, but, at best, they are scattered. Even though the inclinations are lowered in heating the samples from 400 to 500 °C, the site mean inclinations are still higher than for most of the other formations investigated. There may be a stable primary component remaining upon heating to 600 °C, but, if so, it was too weak to measure. The results from the one site in the Alpha member of the Toroweap formation are slightly different from those of Farrell & May (1969) as seen in the following comparison:

	D	I	k	α_{95}	Long.	Lat.	Temp. °C
Present study	154	−20	77	10	117° E	55° N	620
Farrell & May (1969)	157	+ 3	36	7	103° E	47° N	630

The major difference is in the inclination. The inclination of the samples changed markedly with thermal demagnetization, becoming less positive or more negative with each increase in furnace temperature (Fig. 3).

Guadalupian formations

Rocks of Guadalupian age were collected from four formations in Oklahoma, and two formations in New Mexico.

In Oklahoma the collecting was severely restricted by the nature of the rocks in the section. The Flowerpot shale is composed of 70 m of red-brown, gypsiferous shale with many thin beds of orange-brown and greenish-grey, fine-grained sandstone and coarse-grained siltstone units with much gypsum cement. The Blaine formation consists of four named gypsum members interbedded with red shales and dolomite. The Dog Creek shale is composed of 16–20 m of red-brown, silty shale with interbedded gypsum, siltstone, sandstone, and dolomite. The Marlow formation is composed of about 30 m of poorly-cemented, fine-grained, red-to-maroon sandstone with several beds of dolomite. Most were unsuitable for taking either core or oriented block samples. Consequently, the number of sites and the number of samples per site is limited. Two sites were collected in the Flowerpot formation, one coming from the red shale unit which comprises most of the formation and one from the Chickasha member. Only three samples were collected from the Blaine formation. Three sites were collected in the Marlow formation, two of which were from channel-fill deposits, the Doe Creek and Verden members. These were much better indurated than the Marlow formation itself which is so friable that it could be sampled at only one site.

The results from the Chickasha member of the Flowerpot shale have reversed directions, but with high positive inclinations indicating that, although there is a reversed component of magnetization, it is still being masked by a secondary overprinting of normal magnetization after heating to 500 °C. The few samples from the Flowerpot shale had reversed magnetizations, but the samples fell apart when they were heated in the furnace and, therefore, could not be measured. The three samples from the Blaine formation have reversed directions and, although the circle of confidence is large ($\alpha_{95} = 31°$), the palaeopole position (114°E, 51° N) agrees well with the others computed in this study. The Dog Creek shale also gives directions and pole positions that are consistent with the results from the other units studied (Table 1). The directions measured for the samples from the Doe Creek and Verden members of the Marlow formation are scattered, again suggesting a secondary component that has not been eliminated. However, the results from the one site that was collected from the Marlow formation proper are well grouped ($k = 72$, $\alpha_{95} = 7°$) with mean directions and a palaeopole position in good agreement with the results from the Dog Creek and Blaine formations.

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The Guadalupian rocks that were collected in south-eastern New Mexico came from the Seven Rivers and Yates formation as exposed in the back reef area north-west of the Guadalupe Mountains. In this region the Seven Rivers formation consists of about 90 m of mainly gypsum and dolomitic limestone with minor amounts of red and gray gypsiferous shale, siltstones and sandstones. It was sampled at three sites in a stream cut near Carlsbad. The Yates formation in the location sampled consists of interbedded sandstone, dolomite, shale, halite, and gypsum with a total thickness of about 90 m. The only section sampled was in a gulley and only one site (seven samples) was collected. The results from the Seven Rivers formation are consistent with those from the other Guadalupian locations (Table 1). The results from the Yates formation are, however, considerably different. Although the precision parameters are not optimum ($k = 17$, $\alpha_{95} = 17°$), the mean directions do not change much with demangetization, suggesting that the remanent magnetization is stable and the directions are real and that, therefore, they represent a real normal polarity event at the end of Guadalupian time. The computed palaeopole positions for the Seven Rivers and Yates formations, corrected to the northern hemisphere, are as follows:

	Long.	Lat.
Seven Rivers	133° E	50° N
Yates	130° E	54° N

The pole position for the Yates formation is almost antipodal to the pole position for the Seven Rivers formation lending credibility to the suggested interval of normal polarity.

Ochoan formations

Three formations of Ochoan age were sampled in Oklahoma and one in south-eastern New Mexico. None of the rocks sampled were deposited during the first half of the Ochoan epoch.

The formations from which samples were collected in Oklahoma are the Cloud Chief, the Doxey, and the Elk City. Only one site was collected in each of the Cloud Chief and the Doxey formations and four sites were collected at and just above the Doxey–Elk City boundary. The results from the Cloud Chief formation (Table 1), although not conclusive, suggest either another interval of normal polarity during the middle of Ochoan time, a continuation of the normal event found in the Yates formation at the end of Guadalupian time, or possibly the result of strong secondary overprinting. As the sample number is severely limited, it is not possible to determine which is correct. The Doxey and Elk City formations have directions that are very similar to those obtained from other reversely magnetized units through the entire Permian (Table 1). The samples from the Elk City formation were collected from near the bottom and are not correlative with the samples from New Mexico.

Exposures of the Dewey Lake redbeds are limited and, as a result, only one site was collected. The formation is composed primarily of reddish-brown siltstone and shale with an approximate thickness of 60 m. The mean directions and pole position suggest that there was another period of normal polarity at the end of Ochoan time (Table 1).

Discussion of results

The results of this investigation once more confirm the suitability of Permian rocks from the south-western U.S.A. for palaeomagnetic work. The grouping of the computed virtual pole positions (Fig. 4), in general, is even better than previous results suggest. This, and the somewhat higher palaeolatitudes, is almost certainly,

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as Farrell & May (1969) have suggested, the result of more complete elimination of secondary components of magnetization than in the pioneer studies. There is no apparent orderly chronological sequence of pole positions during Permian time.

The thermal demagnetization of all samples has shown that the secondary component of magnetization is seldom easily eliminated. In most cases, temperatures of 500 °C had to be used, and for some, it was found necessary to go as high as 640 °C. For some rocks the secondary component still appeared to be present as the intensity of magnetization decreased to below the measuring limits of the instrument. Experience in demagnetization made it clear that it was legitimate to eliminate from the computations the results from samples with magnetic inclinations greater than 50° as still possessing an appreciable secondary component. This was the only subjective treatment of the demagnetization data.

The most interesting result has been the suggestion that the Kiaman Interval of Irving is broken up by three or possibly four periods of normal magnetization. In view of the fact that most of the redbeds acquired their magnetization chemically (Walker 1967), we are of the opinion that 'magnetic events' in the sense that this expression is used in the Cenozoic are unlikely to be represented and what is observed are epochs which may follow the changing geomagnetic field with an unknown, but possibly appreciable, time lag (that is geomagnetic field changes are not recorded in contemporaneous sediments, but in sediments deposited some unknown time earlier, a time which may even be measurable in millions of years).

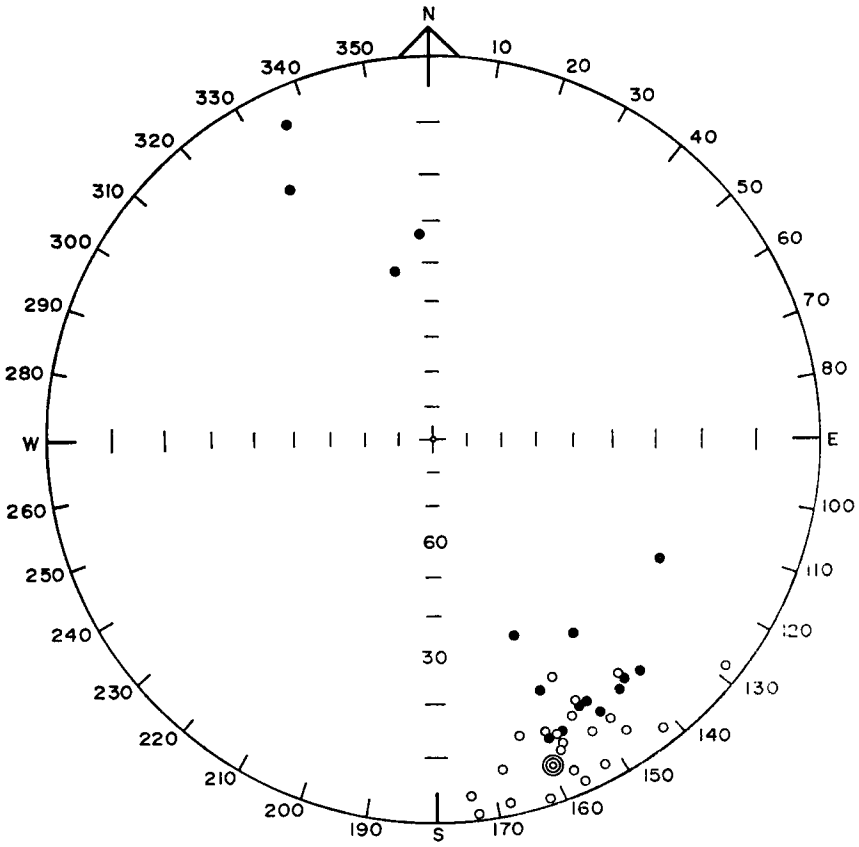


FIG. 4. Stereo plot showing site mean directions for all sites in Table 1.

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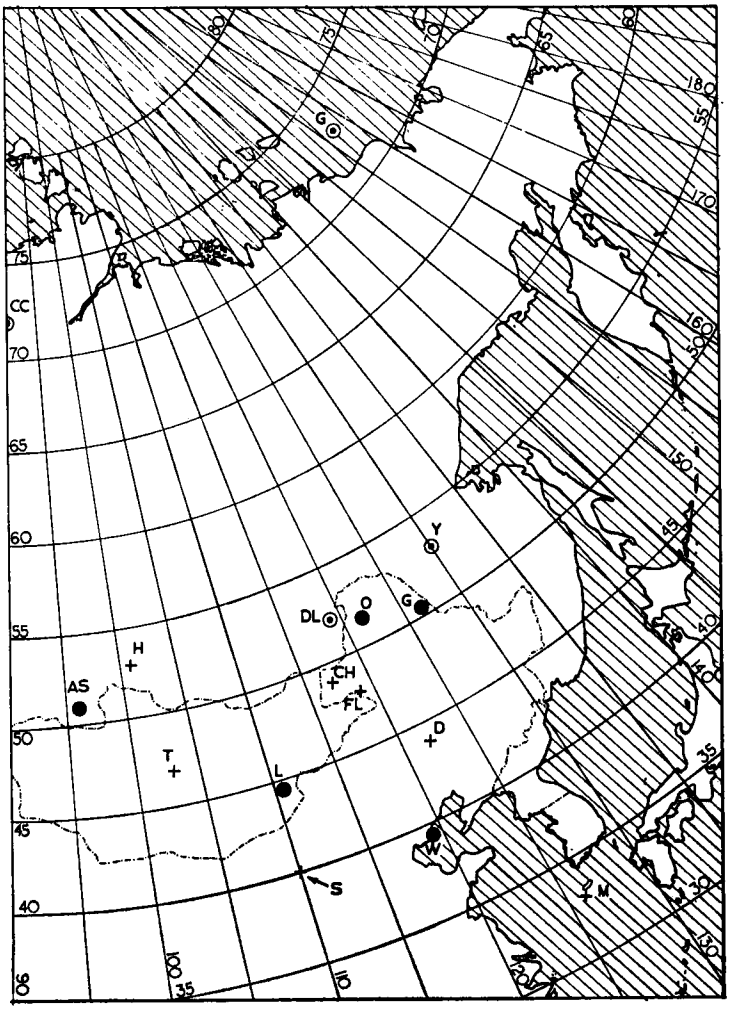


FIG. 5. Map of eastern Asia showing: (1) mean pole positions computed for each epoch during Permian time (black dots labelled W, L, G, and O), (2) the pole positions for the normally magnetized rocks (black circles with dots in them with the letter for the formation name Garber, Yates, Cloud Chief, and Dewey Lake, (3) the Supai results from Oak Creek Canyon, AS, and (4) the previously published Permian results for which the stability had been tested (crosses labelled as follows: CH Halgaito member of the Cutler formation, H for Hermit shale, T for Alpha member of the Toroweap formation (Farrell and May, 1969); S for Supai (Irving, 1964); *FL* for Fountain and Lykens formations and M for Maroon formation (McMahon and Strangway, 1968a and b); and D for Dunkard group (Helsley, 1965)).

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Discounting normal directions of magnetization whose significance in the absence of stability tests is ambiguous, such as Graham (1955) was at pains to point out in some Permian normal directions he had observed, there are few published indications of normal magnetization during Permian times. The best documented are two Tatarian horizons referred to by Kalashnikov (1961) and Khramov (1967), which are paralleled by a single normal horizon recorded in the Werfenian of Slovakia, although here a Triassic age cannot be excluded. It is tempting to regard them as correlating with the youngest normal magnetizations, those of the Dewey Lake and Cloud Chief formations of late and middle Ochoan age. However we do not regard our data as being sufficiently complete to assert the validity of such a correlation. Guicherit (1964) reported a normal magnetization in the middle Permian from samples collected in the Groedener sandstone. However it seems clear that this 100 m sandstone cannot occupy the whole time from the beginning of the Leonardian until the lower part of the Ochoan, and in consequence his result is of doubtful value in attempting to establish a chronology. The Russians also report a normal magnetization at the Carboniferous Permian boundary which does not seem to have been established with certainty elsewhere. However Helsley reported what appears to be a stable normally magnetized NRM from three samples close to the base of the Permian in West Virginia (Helsley 1965).

In Europe most of the sampling has been carried out on the Rotliegendes in which no normal directions of magnetization have, so far, been found. It may therefore be suggested that either the sampling interval was too great, or despite the enormous thickness, the Rotliegendes may represent only the lower part of the Permian section, up to about middle Leonardian of American terminology.

In the American section the normal directions of magnetization are found in the Dewey Lake formation which occurs at the end of Ochoan time and in the Cloud Chief formation of middle Ochoan age. The Yates formation of late Guadalupian age is also normally magnetized, and in the absence of any sampling of formations between the Cloud Chief and Yates formations it remains uncertain whether one or two normal epochs are represented.

A normal direction was also found in the Garber formation of middle Leonardian age, which we regard as the most probable equivalent of Guicherit's Groedener Sandstone result. Below this horizon no normally magnetized rocks have been found. The failure of earlier studies to recognize normal events may probably be attributed to two main causes: the general lack of detailed stratigraphical sampling, with most sampling being restricted to Wolfcampian horizons, and, possibly the mistaken rejection as unstable of rocks with approximately normal directions of magnetization before stability testing became routine procedure. It seems to the authors that there are almost certainly other horizons with normal magnetization awaiting discovery. It is worth recording that no long sequence of normally magnetized rocks was found, the greatest stratigraphical thickness involved was about 50 m.

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